ThoughtFlow: A Negative-Result Falsification Ladder for Sparse-Cache Retention

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Abstract

We present a falsification protocol using repo-local registered successor signals for training-free sparse KV-retention proposals on Mac-local saved sparse-cache fixtures. The contribution is diagnostic, not a positive method: freeze a sparse-cache quality surface, write a signal rule and stop criterion before reading the corresponding saved measurement row, use nominal 0.20-budget proxy rows with achieved keep rates reported as promotion tests and stop signals that fail them, separate stopped-family rows from proxy-baseline rows, and report paired uncertainty plus oracle/headroom diagnostics where those diagnostics are implemented. All benchmark surfaces here are Mac-local saved-trace falsification fixtures, not reasoning-model benchmark claims. As a case study, the original anchor/recent/phase/math family is stopped, recurrence-distance utility (`rdu_topk`) clears one frozen 74-trace surface but fails stricter reproduction, prefix-surprisal utility (`psi_topk`) fails on a fresh 70-trace repo-local C2C-named GSM surface, and value-weighted attention contribution (`vwac_topk`) fails on a fresh 64-trace repo-local C2C-named SVAMP surface. The packet hashes the registration files and measurement artifacts but does not provide an external timestamp service, so we frame the evidence as an auditable repo-local stop ladder. The original FP8 systems target is not part of this claim: the anchor/phase int8 quantization primitive passes Triton interpreter correctness tests against its CPU reference, but no real FP8, CUDA, latency, throughput, or live compression-method claim is made.

1 Contribution

The contribution is a reusable stop-rule protocol, not another sparse-cache policy. The protocol combines seven checks: no-retune frozen surfaces, repo-local registered successor utilities, nominal-budget proxy baselines with achieved keep-rate reporting, stopped-family/proxy-baseline separation, paired uncertainty, oracle/headroom diagnostics for the RDU demotion gates, and a tracked diagnostic packet that hashes the stop artifacts. ThoughtFlow-FP8 is historical naming only: this draft evaluates sparse-cache falsification, not an FP8 serving method. It is the case study showing why each check matters: one plausible signal appears positive on an initial frozen surface, then fails to reproduce under stricter reproduction, while two later signals fail fresh-surface gates outright. If older tracked artifacts say `ALIVE`, `REPRODUCED`, or `PROMOTED`, the current decision manifest supersedes them.

Reviewer takeaway: the protocol is designed to stop attractive false positives before they become GPU or kernel projects. In this case it catches a first-surface `rdu_topk` win, forces same-family and proxy-baseline separation, and then uses independent/fresh saved-trace surfaces to demote or kill every consumed successor without retuning.

The next table is included only to show why the branch looked promising before the fresh gates. It is not a current leaderboard: every candidate policy row in the table is superseded by the demotion ladder that follows.

Protocol check	Purpose	Stop condition
Frozen surface	Prevent post-hoc tuning.	A successor is measured once after its repo-local rule and stop criterion are written.
Nominal budget accounting	Avoid full-cache comparisons.	Must beat R-KV-like and ThinKV-like rows at the registered nominal budget, with achieved keep rates reported rather than assumed equal.
Stopped/proxy split	Separate local family reuse from proxy-baseline robustness.	A stopped-family-only win is diagnostic, not a method claim.
Paired intervals	Avoid mean-only promotion.	For $\Delta = \mathrm{NLL}_{candidate} - \mathrm{NLL}_{baseline}$, promotion requires mean $\Delta \leq -0.03$ and paired CI high below zero versus both proxies.
Oracle/headroom	Show whether a useful policy exists at the budget.	Large oracle gaps block method claims.
Fresh surface	Test robustness.	Failure demotes or kills the signal.

Table 1: Falsification protocol used for each sparse-cache successor.

Signal	Final surface	Final state
anchor family	74-trace frozen sparse-cache gate with nominal-budget ThinKV-like and R-KV-like proxies.	Stopped: no robust nominal-budget win.
`rdu_topk`	Alternate surface plus independent 89-trace saved-trace reproduction.	Killed: first-surface pass fails reproduction.
`psi_topk`	Fresh 70-trace repo-local C2C-named GSM saved-trace surface.	Killed: large NLL regression versus both proxies.
`vwac_topk`	Fresh 64-trace repo-local C2C-named SVAMP saved-trace surface.	Killed: worse than both proxies with paired intervals.

Table 2: Final status matrix. The contribution is the auditable stop protocol and its failure modes, not a live sparse-cache method.

2 Signals Under Test

The original policy attempted to protect anchors and phase labels. The first successor adds a small recent-token reserve. The second successor attempted to protect anchors, reserve half the retained budget for recent tokens, and let phase markers compete with math-state and high-importance reasoning tokens under a saliency bonus. Those policies are now stopped on the current saved traces. The registered `rdu_topk` successor is demoted after stricter reproduction; later `psi_topk` and `vwac_topk` successors are killed by fresh-surface gates.

The proxy rows in Table 3 are deliberately simple nominal 0.20-budget controls with achieved keep rates reported, not faithful reimplementations of the cited systems: `R-KV-like` keeps the first four anchor tokens plus the most recent tail, `ThinKV-like` keeps a recent reserve plus a math-state importance bonus, and `LongFlow-like` is an importance top-k retention rule. The "-like" suffix is part of the local row name: each such row is a Mac-local proxy, not a benchmark comparison to the named method. The local definitions live in `experimental/thoughtflow_fp8/phase2/simulate_phase_retention.py`; the paper uses them only as falsification controls for the stopped signal family.

Policy	Keep rate	NLL	Δ NLL vs full	PPL
full context	1.000	2.101	0.000	9.7
R-KV-like	0.210	3.419	1.319	38.7
ThinKV-like	0.210	3.583	1.482	44.6
LongFlow-like	0.210	3.961	1.861	74.2
ThoughtFlow	0.210	3.961	1.861	74.2
ThoughtFlow-recent	0.210	3.562	1.461	44.2
ThoughtFlow-saliency-recent	0.210	3.434	1.333	38.1
Held-out R-KV-like	0.211	3.482	–	43.7
Held-out sweep best	0.211	3.480	–	43.7

Table 3: Case-study retained-context continuation proxy on distilgpt2 traces. Lower NLL is better. R-KV-like, ThinKV-like, and LongFlow-like rows are Mac-local proxy baselines in this harness, not faithful implementations. Full context is an uncompressed reference row, not a nominal-budget compressed baseline. The held-out sweep row is selected on 12 train traces and reported on 12 held-out traces, so its NLL is not directly averaged with the all-trace rows.

3 Hidden/KV Telemetry

We then ran a Mac-local telemetry gate using real distilgpt2 tensors rather than only text markers. The probe ranks retained tokens by attention received, final-hidden norm, key norm, value norm, or combined KV norm. This is still not sparse-KV decoding, but it checks whether the interpretable phase policy is merely preserving labels that hidden/KV saliency would not select.

Policy	Keep rate	Anchor	Phase	Math-state
ThoughtFlow	0.207	1.000	1.000	0.918
LongFlow-like	0.207	1.000	1.000	0.918
ThinKV-like	0.207	1.000	0.948	0.848
`value_norm_topk`	0.207	0.000	0.492	0.845
train-selected sweep	0.207	1.000	0.430	0.956

Table 4: Hidden/KV saliency telemetry on 24 distilgpt2 traces. The strongest real-saliency proxy by phase+math recall is `value_norm_topk`. ThoughtFlow has higher phase recall than it, but LongFlow-like ties ThoughtFlow under the local importance heuristic.

Paired against `value_norm_topk`, ThoughtFlow's phase-recall delta is +0.508 with 95% CI [+0.436, +0.579]. The math-state delta is only +0.073 with 95% CI [-0.078, +0.223]. This rules out a method claim based on phase recall alone: it may be marker preservation rather than useful cache retention.

4 Sparse-Cache Case Study Gate

We next ran a CPU-only sparse-cache probe. The model processes the full prefix once, each policy prunes the returned KV cache to the same 0.20 budget, and the continuation is scored from the pruned cache. This is not a Triton, CUDA, FP8, latency, or throughput result. It is closer to the cache-use failure mode than retained-text NLL, but still CPU-only diagnostic evidence.

A bounded sparse-cache sweep over 24 nearby anchor/recent/phase/math configurations selects `tf_sparse_r0.55_p0.05_m0.12_a2` on the 12 train traces. On the 12 held-out traces it gets NLL 3.340 versus ThinKV-like 3.385 and R-KV-like 3.420. This clears the mean 0.03 margin, and the paired delta versus R-KV-like is -0.080 with 95% CI [-0.152,-0.014]. However, the paired delta versus ThinKV-like is -0.045 with 95% CI [-0.226,+0.182]. The fixed ThoughtFlow-saliency-recent incumbent has lower held-out mean NLL, 3.304, but its paired CIs also cross zero. This is a promising falsification candidate, not a method result.

Policy	Keep rate	NLL	Paired Δ vs R-KV-like
full cache	1.000	2.142	-1.296 [-1.533,-1.066]
ThoughtFlow-saliency-recent	0.210	3.372	-0.067 [-0.151,+0.011]
ThinKV-like	0.210	3.389	-0.049 [-0.192,+0.077]
ThoughtFlow-recent	0.210	3.399	-0.040 [-0.169,+0.074]
ThoughtFlow sweep best	0.210	3.436	-0.002 [-0.007,+0.000]
R-KV-like	0.210	3.438	0.000
LongFlow-like	0.210	3.588	+0.150 [-0.016,+0.300]
ThoughtFlow	0.210	3.588	+0.150 [-0.006,+0.297]

Table 5: CPU sparse-KV continuation quality on 24 distilgpt2 traces. R-KV-like, ThinKV-like, and LongFlow-like rows are Mac-local proxy baselines, not faithful implementations. Negative paired delta is better than R-KV-like. ThoughtFlow-saliency-recent is best in mean NLL, but it clears the strongest non-ThoughtFlow row by only 0.017 NLL and remains below the registered 0.03 promotion margin.

We then froze `ThoughtFlow-saliency-recent` and `tf_sparse_r0.55_p0.05_m0.12_a2` and ran a larger 74-trace sparse cache slice with no retuning. ThinKV-like is the best compressed row among the stopped family comparison at NLL 3.900. The frozen sparse ThoughtFlow row gets 3.908, with paired delta -0.031 versus R-KV-like and 95% CI [-0.078,+0.020], but +0.008 versus ThinKV-like with 95% CI [-0.060,+0.085]. ThoughtFlow-saliency-recent gets 3.920. The larger no-retuning gate therefore stops this policy family.

After that stop decision, we registered `rdu_topk` in the repo-local stop ladder. It computes recurrence-distance utility from full-prefix prefill attentions using lag buckets [8,15], [16,31], [32,63], and [64,∞], then keeps the top budget by utility with ties broken by lower position. It does not use continuation tokens, trace labels, recent reserves, anchors, phase markers, or math-state bonuses. After `rdu_topk` failed reproduction, we registered `psi_topk` (high self-surprisal prefix tokens) and `vwac_topk` (future attention weighted by cached value-vector norm).

5 Evidence Ladder

The independent saved-trace check rules out claiming `rdu_topk` as a robust method on the available surfaces. A coarse trace-level decomposition explains where the reproduction fails: `rdu_topk` is +0.213 NLL worse than R-KV-like on long-prefix/high-RDU-density rows but -0.049 better on short-prefix/low-density rows. The large-regression buckets are exactly the buckets a robust policy would need to handle before any long-context claim.

Prefix-surprisal successor. A later consumed successor tested whether retaining high self-surprisal prefix tokens could preserve information that is hard to reconstruct. The signal uses only prefill logits, not continuation loss, trace labels, recent reserves, or RDU lag buckets. On a fresh 70-trace distilgpt2 C2C GSM saved-trace surface, `psi_topk` fails decisively: ThinKV-like is best compressed at NLL 3.906, R-KV-like reaches 3.960, and `psi_topk` reaches 7.899. Paired deltas are +3.939 [+3.720,+4.144] versus R-KV-like and +3.994 [+3.773,+4.205] versus ThinKV-like, where positive is worse for `psi_topk`. This rules out this exact signal in the current Mac sparse-cache harness. Achieved keep rates are 0.2199 for R-KV-like and 0.2156 for both `psi_topk` and ThinKV-like.

Value-weighted attention successor. The next consumed successor tested a cache-state utility: future prefix attention to a token multiplied by the token's cached value-vector norm. On a fresh 64-trace distilgpt2 C2C SVAMP saved-trace surface, `vwac_topk` also fails. R-KV-like is best compressed at NLL 4.096, ThinKV-like reaches 4.162, and `vwac_topk` reaches 4.336. Paired deltas are +0.241 [+0.030,+0.530] versus R-KV-like and +0.174 [+0.002,+0.392] versus ThinKV-like. Achieved keep rates are 0.2389 for R-KV-like and 0.2219 for both `vwac_topk` and ThinKV-like.

Gate	Readout	Decision
Synthetic retention	Anchor/phase labels can be protected at matched keep rate.	Intuition only.
Retained-text NLL	Original ThoughtFlow rows do not beat the strongest proxy.	Weakened.
Hidden/KV telemetry	Phase recall improves, but math-state impact is not stable and LongFlow-like ties the heuristic.	Diagnostic.
CPU sparse-cache probe	A tuned sparse ThoughtFlow row is promising in mean but not robust against ThinKV-like uncertainty.	Not promoted.
Frozen 74-trace gate	`rdu_topk` beats ThinKV-like and R-KV-like on the first frozen surface.	Candidate only.
Same-slice rerun	Cached result reproduces exactly on the deterministic Mac stack.	Repro bookkeeping.
Alternate surface	Cross-family rows still lose, but a stopped same-family row beats `rdu_topk` by 0.006 NLL.	Weakened.
Independent saved traces	R-KV-like is best compressed at NLL 3.981; `rdu_topk` reaches 4.014.	Killed.
Prefix-surprisal fresh saved-trace surface	On distilgpt2 C2C GSM saved traces, `psi_topk` reaches NLL 7.899 versus ThinKV-like 3.906 and R-KV-like 3.960.	Killed.
Value-weighted attention fresh saved-trace surface	On distilgpt2 C2C SVAMP saved traces, `vwac_topk` reaches NLL 4.336 versus R-KV-like 4.096 and ThinKV-like 4.162.	Killed.

Table 6: ThoughtFlow-FP8 evidence ladder. The first-surface pass is shown as one step that failed to reproduce, not as the headline result.

Registration	SHA-256 prefix	Measurement artifact
`rdu_topk`	`c49da7df`	Same-surface pass, alternate-surface failure, and independent-trace failure are bundled in the diagnostic packet.
`psi_topk`	`cd8a6a10`	One-shot fresh distilgpt2 C2C GSM saved-trace sparse-cache check, artifact SHA prefix `d0a08035`.
`vwac_topk`	`a701495b`	One-shot fresh distilgpt2 C2C SVAMP saved-trace sparse-cache check, artifact SHA prefix `80a35954`.

Table 7: Registration-to-measurement ledger. Full hashes and artifact paths are recorded in the tracked diagnostic packet; this table exposes the repo-local audit chain inside the paper. The packet does not independently prove external timestamp ordering.

6 Claim Boundary

Limitations. The ladder is demonstrated on Mac-local saved traces and CPU sparse-cache scoring, not native serving. The trace families are useful falsification surfaces, but they do not establish broad task coverage. The stopped signals also do not rule out future sparse-cache utilities; they only rule out the registered policies under the reported surfaces and budgets. The scored surfaces use `distilgpt2` saved continuations and saved prefix/KV telemetry as falsification fixtures; they are not reasoning-model benchmarks and not faithful external implementations of ThinKV, R-KV, LongFlow, LazyEviction, ForesightKV, PM-KVQ, or KVQuant.

Historical row	Traces	Keep	NLL	Δ vs ThinKV	Later decision
full cache	74	1.000	2.848	-1.052 [-1.199,-0.919]	reference only
rdu_topk	74	0.213	3.779	-0.121 [-0.211,-0.037]	reproduction kill
ThinKV-like	74	0.213	3.900	0.000	proxy baseline
frozen sparse TF	74	0.213	3.908	+0.008 [-0.060,+0.085]	stopped family
TF-saliency-recent	74	0.213	3.920	+0.020 [-0.030,+0.074]	stopped family
R-KV-like	74	0.214	3.939	+0.039 [-0.015,+0.099]	proxy baseline
LongFlow-like	74	0.213	4.158	+0.258 [+0.178,+0.337]	proxy baseline

Table 8: Historical failed-to-reproduce one-shot frozen sparse-cache successor evaluation. R-KV-like, ThinKV-like, and LongFlow-like rows are Mac-local proxy baselines, not faithful implementations. rdu_topk also beats R-KV-like by -0.160 NLL with 95% CI [-0.264,-0.050] on this first surface, but later alternate-surface and independent-trace gates fail to reproduce the promotion.

rdu_topk check	Readout	Decision
First frozen surface	74 traces, NLL 3.779 versus ThinKV-like 3.900.	First-surface pass.
Same-slice rerun	Exact deterministic reproduction; 0.145 NLL above compressed oracle.	Bookkeeping only.
Alternate surface	Same rule, NLL 3.594; a stopped same-family row reaches 3.588.	Weakened.
Independent traces	89 SVAMP traces; R-KV-like 3.981, rdu_topk 4.014.	Killed.
Failure bucket	+0.213 NLL on long-prefix/high-RDU-density rows.	Mechanism target.

Table 9: rdu_topk demotion ladder. The first-surface pass is preserved as evidence that the protocol catches tempting positives, not as a live method claim.

7 Stop Rule and Reopen Gate

The branch now contributes a diagnostic falsification ladder. The stopped anchor/recent/phase/math policy family remains ruled out as a tunable branch on this trace set, rdu_topk fails reproduction, and psi_topk/vwac_topk fail their fresh-surface gates. Any future positive-method claim requires a genuinely new registered utility signal evaluated once on a fresh/larger frozen surface with strict stopped-family, proxy-baseline, and oracle/headroom reporting. The Triton interpreter gate is no longer the blocker for the quantization scaffold; method evidence is. This means no real FP8, CUDA, latency, or throughput result is claimed.

The methodological value of the artifact is procedural: it reduces the chance that a plausible sparse-cache heuristic is promoted from a narrow frozen slice into implementation work before robustness exists. The stopped rows are therefore not failed leaderboard entries; they are worked examples of what the ladder is meant to prevent. The intended contribution is a reusable gate for future proposals, not a leaderboard entry for the stopped policies.

8 Related Work and Scope

This draft intentionally does not claim a new state-of-the-art KV-compression method. ThinKV, LongFlow, R-KV/R-KVHash-style retention, LazyEviction/ForesightKV-style future-use ideas, PM-KVQ, KVQuant, TriAttention, KVzip/KVzap, Q-Filters, and KV-Direct already occupy much of the sparse-cache, long-reasoning compression, and quantized-cache design space ThinKV (2026); LongFlow (2026); R-KV (2025); LazyEviction (2025); ForesightKV (2026); PM-KVQ (2025); KVQuant (2024); Recent KV-cache methods (2025–2026). Our contribution is narrower: a Mac-local falsification ladder that shows how plausible interpretable retention signals can clear one frozen surface, fail stricter reproduction, or collapse on fresh surfaces. The project name retains "FP8" because that was the original systems target, but the current artifacts only validate CPU sparse-cache scor-

Lesson	Observed failure	Protocol implication
First-surface positives are weak evidence.	`rdu_topk` passed one frozen slice, then failed same-family and independent surfaces.	Require fresh-surface reproduction before promotion.
Density regimes matter.	Long-prefix/high-RDU-density rows regress by +0.213 NLL.	Bucket diagnostics should be mandatory, not post-hoc color.
Oracle gaps prevent overclaiming.	Same-slice reproduction remained 0.145 NLL above the compressed oracle.	Report headroom before adding implementation work.
Consumed signals should stay consumed.	`psi_topk` and `vwac_topk` failed one-shot gates.	Do not retune failed utility families on the same traces.

Table 10: Methodological lessons from the stopped sparse-cache ladder.

Claimed	Not claimed
Preregistered sparse-cache falsification ladder.	State-of-the-art KV compression.
Nominal-budget Mac sparse-cache scoring with achieved keep rates reported.	Native FP8 serving quality.
Stopped-family/proxy-baseline and fresh-surface stop rules.	CUDA kernel speed, latency, throughput, or HBM savings.
Triton-interpreter parity for an int8 anchor/phase scaffold primitive.	A live deployable compression method or FP8/cache-serving correctness.

Table 11: Boundary between the workshop claim and the stopped systems-method target.

ing and an int8/Triton-interpreter reference primitive; they do not measure FP8 numerical drift or native FP8 serving. We do not benchmark these external systems; they define the crowded method space that motivates narrowing this paper to stop-rule evidence. The local artifact filenames use `c2c` as a repo shorthand for saved cross-condition trace fixtures. They are not Cache-to-Cache communication runs: Cache-to-Cache projects and fuses one model's KV cache into another with a learned gate Cache-to-Cache (2025), while this draft only prunes cached positions inside a single distilgpt2 continuation fixture. The contribution is an audited domain-specific case study of sparse-cache stop rules, not a general claim that preregistration or negative-results methodology is novel.

9 Reproducibility

All reported gates use tracked scripts under `experimental/thoughtflow_fp8/phase2/` and the repo-local `./venv_arm64`. The frozen sparse-cache probe, same-slice rerun, alternate-surface rerun, independent saved-trace rerun, and fresh-surface successor checks each write Markdown and JSON artifacts with promotion criteria, paired intervals, and no-retuning decisions; oracle/headroom fields are present on the RDU demotion gates and are not claimed for the PSI/VWAC fresh successor packets. The tracked diagnostic packet hashes the saved JSON artifacts and records available model/source metadata plus input hashes for each saved artifact where those inputs are explicit. Legacy default trace paths are labeled as inferred from `run_real_trace_retention.DEFAULT_TRACES`. Its builder refuses regeneration when `experimental/thoughtflow_fp8/` is dirty, and the current manifest records a clean path status at generation plus a clean status for hashed input paths. The hashed repo-local C2C-named fresh-surface inputs are tracked in the repository despite the root results ignore rule. The current package should still be read as a machine-checked falsification note rather than a full artifact-evaluation release. The stable local correctness command is:

```
./venv_arm64/bin/python \
  experimental/thoughtflow_fp8/phase2/build_diagnostic_packet.py \
```

```
  --output .debug/thoughtflow_diagnostic_packet_check
```

Then run:

```
TRITON_CPU_BACKEND=1 TRITON_INTERPRET=1 \
TRITON_HOME="$PWD/.debug/triton_home" \
./venv_arm64/bin/python -m pytest \
  experimental/thoughtflow_fp8/phase2/tests \
  experimental/thoughtflow_fp8/phase4/tests -q -rs
```

Gate artifact	Script	Input surface	Hash prefix
frozen probe	`frozen_sparse_cache_probe.py`	frozen mixed traces, 74 scored	`f79c286c`
same-surface RDU	`rdu_no_retune_reproduction_check.py`	same frozen traces, 74 scored	`9a6420cf`
alternate RDU	`rdu_alt_surface_reproduction_check.py`	alternate measured surface, 73 scored	`1302ce31`
independent RDU	`rdu_independent_trace_reproduction_check.py`	saved chat traces, 89 scored	`5a568e1c`
`psi_topk`	`psi_fresh_sparse_cache_check.py`	distilgpt2 C2C GSM, 70 scored	`d0a08035`
`vwac_topk`	`vwac_fresh_sparse_cache_check.py`	distilgpt2 C2C SVAMP, 64 scored	`80a35954`

Table 12: Diagnostic-packet command map. Replay entry points, source metadata, and saved-artifact SHA-256 values are in the tracked manifest. All rows are saved-trace proxy fixtures, not faithful LongFlow, ThinKV, or R-KV benchmark results.

10 Artifacts

Key artifacts are the current decision manifest, tracked diagnostic provenance packet, and reviewer pack: `experimental/thoughtflow_fp8/phase2/current_decision_manifest_20260506.md`, `experimental/thoughtflow_fp8/phase2/diagnostic_packets/thoughtflow_diagnostic_packet_20260506/manifest.json`, and `experimental/thoughtflow_fp8/paper/reviewer_pack.md`. The diagnostic packet hashes the saved falsification JSON artifacts and registration files; README, progress, reviewer pack, and the repo-local Triton interpreter note at `experimental/thoughtflow_fp8/phase4/triton_interpreter_note_20260506.md` are separate handoff artifacts.

References

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KVQuant: Towards 10 Million Context Length LLM Inference with KV Cache Quantization. arXiv:2401.18079, 2024. https://arxiv.org/abs/2401.18079.

Cache-to-Cache: Direct Semantic Communication Between Large Language Models. arXiv:2510.03215, 2025. https://arxiv.org/abs/2510.03215.

Recent KV-cache methods include TriAttention (arXiv:2604.04921), KVzip (arXiv:2505.23416), KVzap (arXiv:2601.07891), Q-Filters (arXiv:2503.02812), and KV-Direct (arXiv:2603.19664).